

Spreading 43 Points in the Plane

Limin Ge*

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Abstract

We place 43 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 40.63406+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 44-point configuration gives 41.63370; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 43 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 40.63406\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 43$ is not on the page. Deleting a point from our 44-point configuration gives 41.63370; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n43.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
40634064449975573951068738
denominator:
999999999998783223429397
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

43 points, x and y per line. File: friedman-maxmin2d/coords/n43.txt.

```
-10.487893317983 6.267382044272
-6.506885481135 7.286912806333
-9.348548256518 2.818689680962
-6.305542041738 5.632341336155
-6.270854600891 4.511156887431
-7.458810926347 3.093742740249
-7.156598951622 4.046983487559
-8.757598573138 5.236382062020
-5.290562517858 4.708710505322
-9.688594851055 6.868316284242
-7.728270013830 1.887672700699
-7.356831334314 7.813782853559
-9.586837039005 5.833679862915
-6.471794220048 3.254360365330
-10.510865097134 4.289186912549
-8.755835307506 7.228815429757
-5.800425862801 6.579159454054
-9.674618243692 4.837540083515
-9.796516802822 1.924640363797
-7.765024718008 5.806669449500
-8.669817368451 6.232521841420
-6.826203629582 2.319270011175
-10.651319051614 2.443594230206
-10.238879998804 7.703293084020
-7.115712732307 5.046147296486
-11.644436080634 5.220916712064
-5.609622107671 3.760975849341
-11.038178465731 7.102358844050
-11.509118791186 4.230114395163
-8.307866761202 8.122864746921
-6.954854027439 6.392863489169
-9.367103651867 3.885996718336
-11.486147012036 6.208309526886
-9.306120455254 8.063792229535
-8.360877310595 2.662145429773
-8.808845856899 1.768096112609
-5.308450189345 5.708550508127
-8.001457083038 4.581973896615
-8.392596043431 3.661642264179
-10.203350505310 3.337643547371
-7.804799880618 6.919733536395
-10.646182386581 5.279989229450
-11.201604199362 3.278571029984
```